

COMPUTER NETWORKS

Week-6

Network Communication

- Communication involves at least two entities
- One that **sends** information
- Another that **receives** information
- Most packet switching communications systems contain intermediate entities (i.e., devices that forward packets)
- For successful communication, all entities in a network must agree on how information will be represented and communicated
- For example, when two electronic entities communicate over a wired network, both sides must agree on the voltages to be used and the format of messages

Interoperability

- The ability of two entities to communicate without any misunderstandings → correct interoperability
- Communication protocol, network protocol, or protocol → some specifications for network communication
- Protocol → a procedure to be followed during an exchange
- Another aspect of a protocol → define the appropriate action to take for each possible abnormal condition
- E.g. an error or unexpected condition occurs, a response is expected, but no response arrives

Protocol Suites

- Protocols must be both complete and efficient
- To avoid duplication of effort, each protocol should handle a part of communication not handled by other protocols
- → protocols are designed in complete, cooperative sets called suites or families
- Each protocol in a suite handles one aspect of communication
- The entire suite is designed to allow the protocols to work together efficiently

Layering Models

- The fundamental abstraction used to collect protocols into a unified whole → a layering model
- A layering model describes how all aspects of a communication problem can be partitioned into pieces that work together
- Each piece = a **layer**
- Protocols in a suite are organized into a linear sequence
- This helps both protocol designers and implementors to concentrate on one aspect of communication at a given time

The Network Layer

- Responsible to take packets from the source to the destination
- Getting to the destination may require making many hops at intermediate routers along the way
- Network layer is the lowest layer that deals with end-to-end transmission

Network Layer

- It must know about the **topology** of the network (i.e., the set of all routers and links) and choose appropriate **paths**
- It has to avoid overloading some of the communication lines and routers while leaving others idle (traffic control)
- It is also responsible to resolve the issues when the source and destination are in different networks

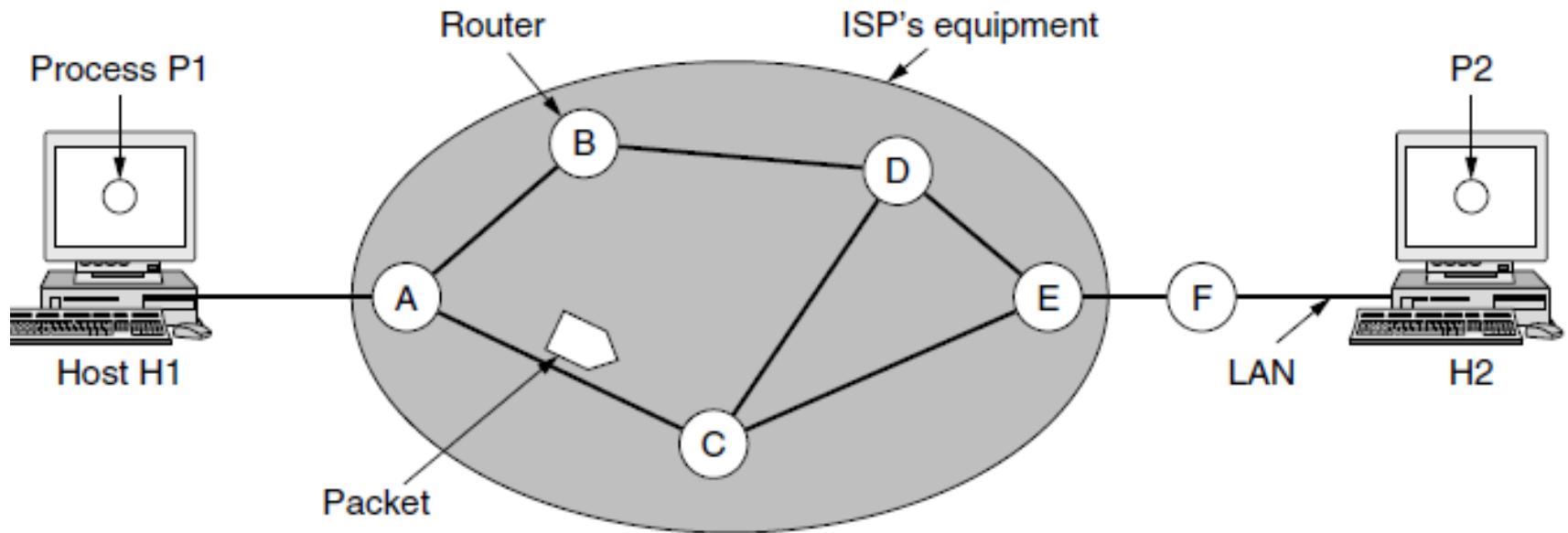
Store-and-Forward Packet Switching

- The goal of a WAN is to allow as many computers as possible to send packets simultaneously
- The fundamental paradigm used to achieve simultaneous transmission is known as *store and forward*
- A switch *buffers* packets in memory

Store & Forward Paradigm

- The *store* operation occurs when a packet arrives:
- I/O hardware inside the switch places a copy of the packet in memory
- The *forward* operation occurs once a packet has arrived and is waiting in memory
- The processor examines the packet, determines its destination, and sends the packet to the destination

Example-1



Example-1

- Host *H1* is directly connected to the ISP's router *A*
- Host *H2* is on a LAN, which might be an office Ethernet, with a router, *F* {owned by the customer}
- A host can transmit a packet to the nearest router, either on its own LAN or over a point-to-point link to the ISP
- The packet is stored there until it has fully arrived and the link has finished its processing by verifying the **checksum**
- Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered

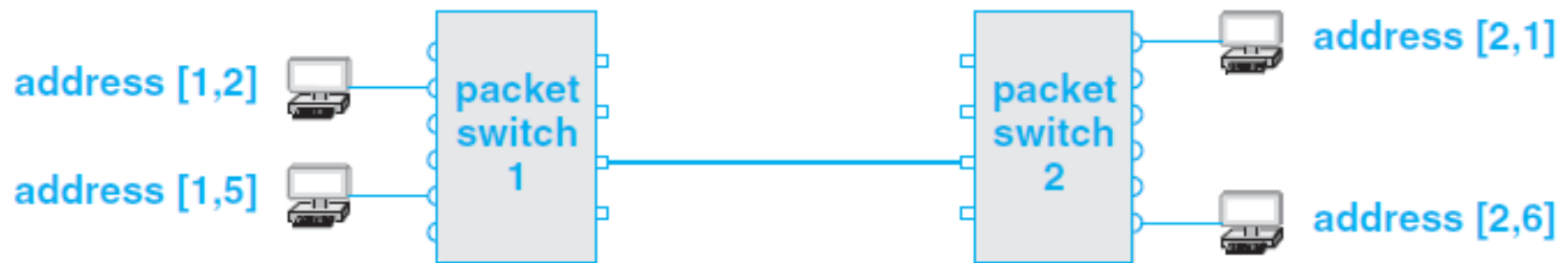
Switches & Routers

- Devices (PCs, phones, printers) connect to a switch
- The switch connects to a router
- The router connects to the Internet
- Switch → connects devices within the same network (LAN)
- Router → connects different networks (LAN to LAN or LAN to Internet)

Addressing in a WAN

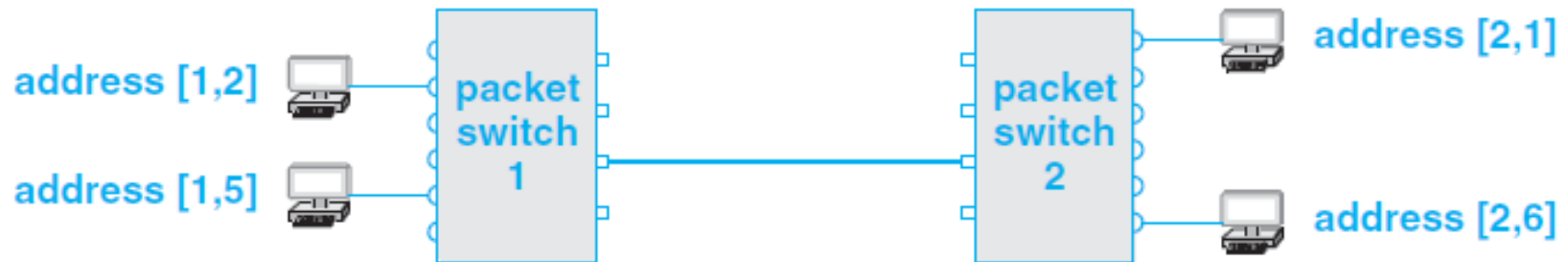
- WAN's addresses follow a key concept that is used in the Internet:
- *hierarchical addressing*
- This divides each address into two parts:
(site, computer at the site)
- Just as we have been talking about the prefix and suffix

Hierarchical Addressing



To find out the total number of ports on a CISCO switch, run the command `show ip interface brief`

Hierarchical Addressing



- (switch #, Port#)
- Each Address is a pair of decimal integers
- A computer connected to port 6 on packet switch 2 is assigned address [2,6]
- As a single binary value, initial bits of the binary value represent a packet switch number and the remaining bits identify a computer

Next-Hop Forwarding

- When a packet arrives, a packet switch must choose an outgoing path over which to forward the packet
- If a packet is destined for a local computer, the switch sends the packet directly to the destination
- Otherwise, the packet must be forwarded over one of the connections that leads to another switch

Next-Hop Forwarding

- Software in the packet switch uses the destination address in the packet to make the decision
- The software extracts the packet switch number from the address
- If the extracted value is identical to the packet switch's own ID, the packet is intended for a computer on the local packet switch
- Otherwise, the packet is intended for a computer on another packet switch
- Algorithm 18.1 may be referred

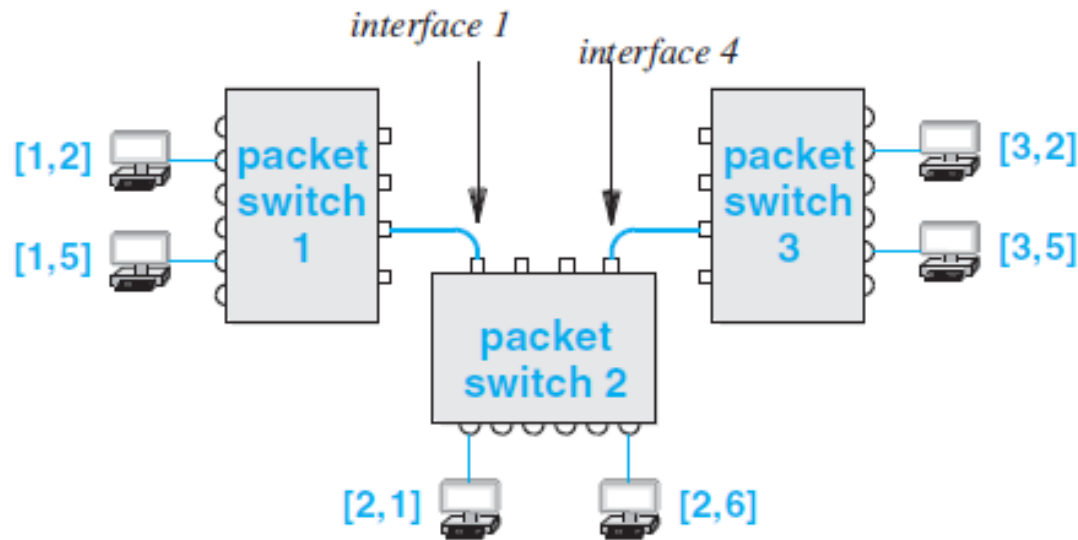
Next-Hop Forwarding

- So a packet switch does not need to keep complete information about how to reach all possible computers
- It does not need to compute the entire route a packet will follow through the network
- A switch only needs to know which outgoing link to use to reach a given switch
- → compute the *next hop* for a packet

Next-Hop Forwarding

- Each packet switch contains a *forwarding table* that lists all possible packet switches and gives a next hop for each

Expl-2 (next hop forwarding table for S2)



to reach	send to
switch 1	interface 1
switch 2	local delivery
switch 3	interface 4

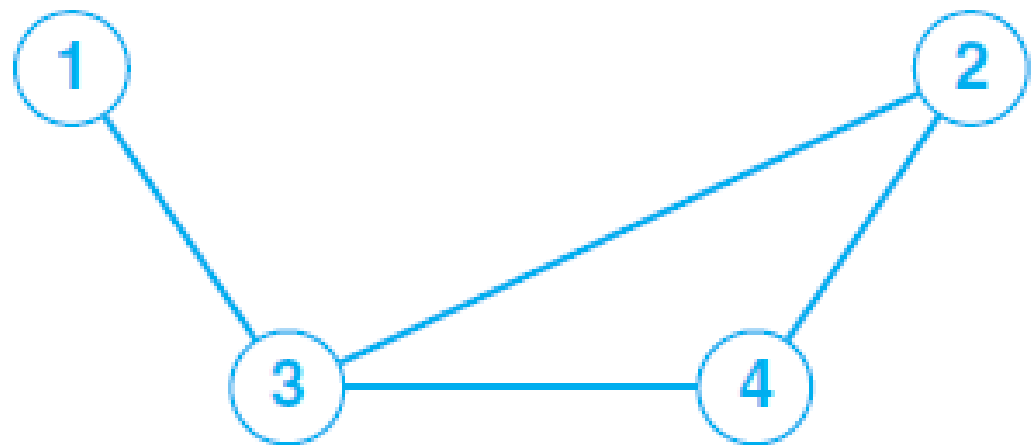
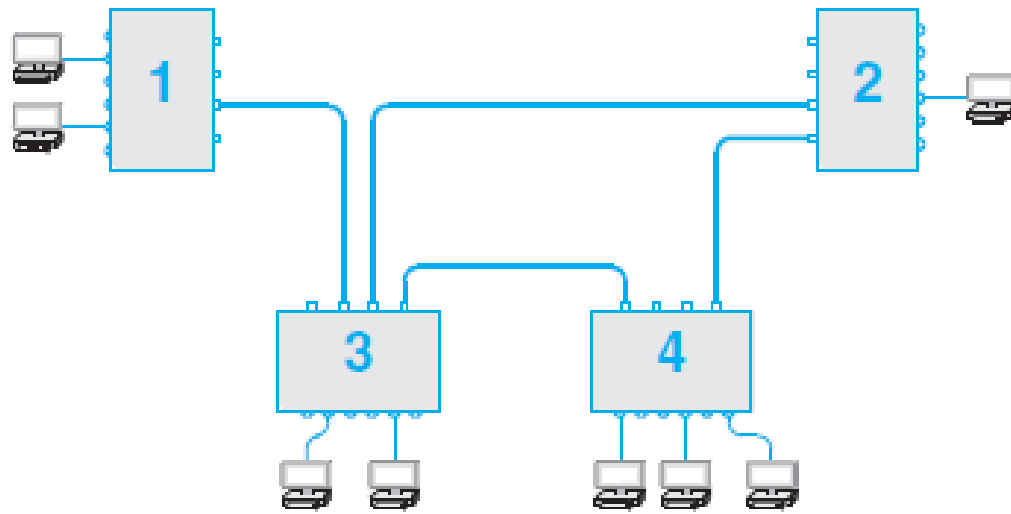
Example-2

- (switch#, port#)
- If a packet is destined for $[3,5]$, the switch extracts 3, consults the table, and forwards the packet to interface 4, which leads to switch 3
- A switch uses only the first part of the destination address until the packet reaches the final switch
- Then switch uses the second part of the address to choose a specific computer

to reach	send to
switch 1	interface 1
switch 2	local delivery
switch 3	interface 4

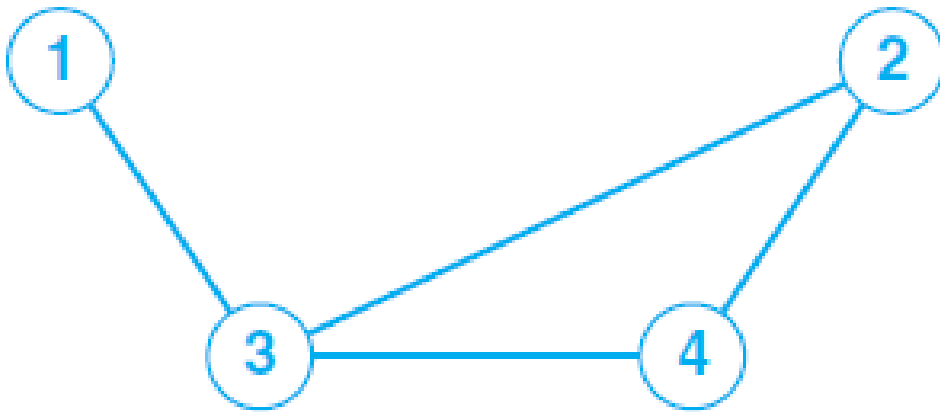
Computing path in a WAN

- The easiest way to think about route computation in a WAN is to think of a graph that models the network
- The software uses the graph to compute the shortest path to all possible destinations
- Each *node* in the graph corresponds to a switch in the network (individual computers are not part of the graph)
- If the network contains a direct connection between a pair of switches, the graph contains an *edge* or *link* between the corresponding nodes



Calculating Forwarding Table

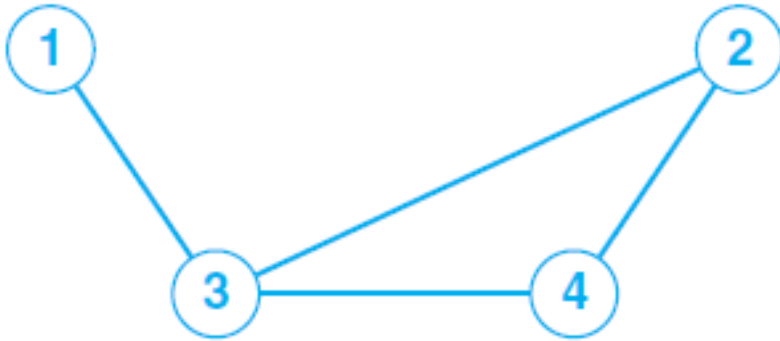
- For node 1
- The notation (k, j) denotes a link from node k to node j



to reach	next hop
1	—
2	(1,3)
3	(1,3)
4	(1,3)

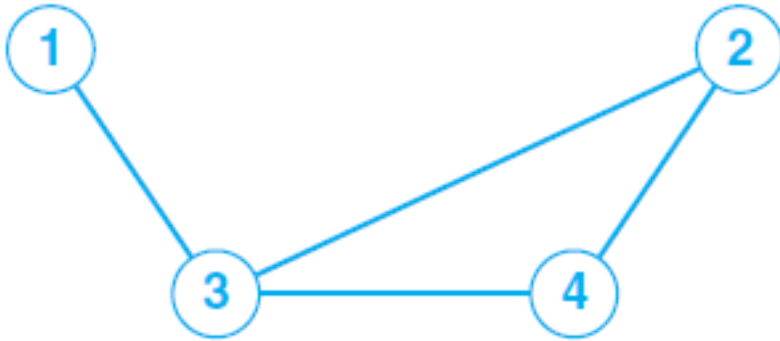
Node 1

For Node-2



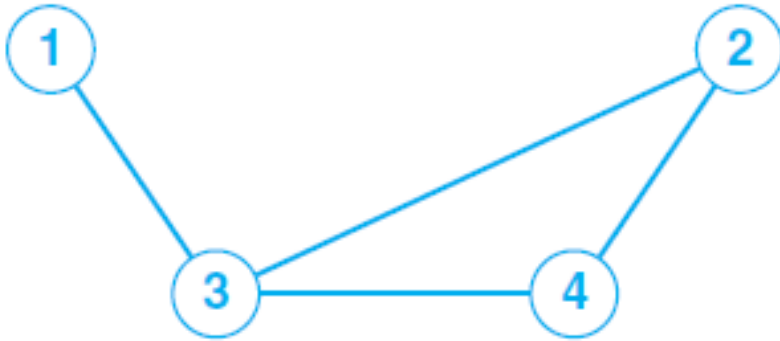
To Reach	Next Hop
1	(2,3)
2	
3	
4	

For Node-2



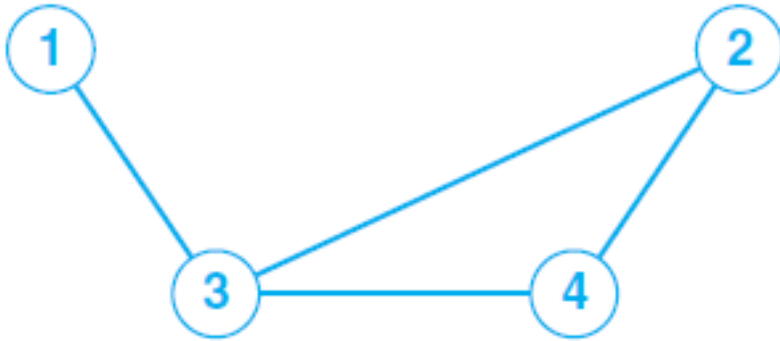
To Reach	Next Hop
1	(2,3)
2	-
3	(2,3)
4	(2,4)

For Node-3 (complete)



To Reach	Next Hop
1	
2	
3	
4	

For Node-4



To Reach	Next Hop
1	
2	
3	
4	

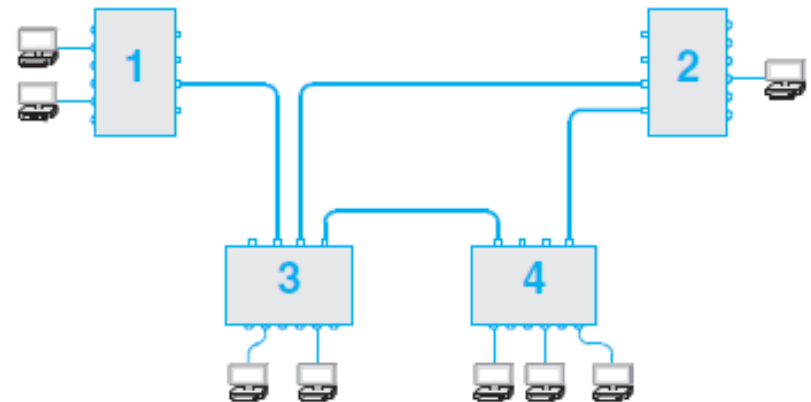
Defaults Routes

to reach	next hop
1	—
2	(1,3)
3	(1,3)
4	(1,3)

Node 1

- The table contains many entries that point to the same next hop
- the packet switch has only one connection to the network

except for the entry that corresponds to the node itself, all entries have a next hop that points to the link from Node 1 to Node 3.



Default Routes

- Most WAN systems include a mechanism that can be used to eliminate the duplicate entries
- → *default route*
- Only one default entry is allowed in a forwarding table, and the entry has lower priority than other entries

to reach	next hop
1	—
2	(1,3)
3	(1,3)
4	(1,3)

Node 1

to reach	next hop
1	—
*	(1,3)

Calculate the rest of the tables

- Node2
- Node3
- Node4

Node-2

to reach	next hop
1	(2,3)
2	—
3	(2,3)
4	(2,4)

Node 2

Node-2

to reach	next hop
1	(2,3)
2	—
3	(2,3)
4	(2,4)

Node 2

to reach	next hop
2	—
4	(2,4)
*	(2,3)

Node-3

to reach	next hop
1	(3,1)
2	(3,2)
3	—
4	(3,4)

Node 3

Node-3

to reach	next hop
1	(3,1)
2	(3,2)
3	—
4	(3,4)

Node 3

to reach	next hop
1	(3,1)
2	(3,2)
3	—
4	(3,4)

Node 3

Node-4

to reach	next hop
1	(4,3)
2	(4,2)
3	(4,3)
4	—

Node 4

Node-4

to reach	next hop
1	(4,3)
2	(4,2)
3	(4,3)
4	—

Node 4

to reach	next hop
2	(4,2)
4	—
*	(4,3)

Forwarding

- When a packet arrives at a router's input link, the router must move the packet to the appropriate output link

Forwarding vs Routing

- **Forwarding** refers to the router-local action of transferring a packet from an input link interface to the appropriate output link interface
- Forwarding takes place at very short timescales (typically a few nanoseconds)
- Forwarding is typically implemented in hardware
- **Routing** refers to the network-wide process that determines the end-to-end paths that packets take from source to destination
- Routing is often implemented in software

Example-1

Routing

- Bank system decides:
 - Transfer to another bank → go through central clearing system
 - International transfer → use SWIFT network

Forwarding

- Each system node:
 - Sends transaction to next system
 - Processes based on instructions
- Routing = Which path should money take?
Forwarding = Send it to the next system

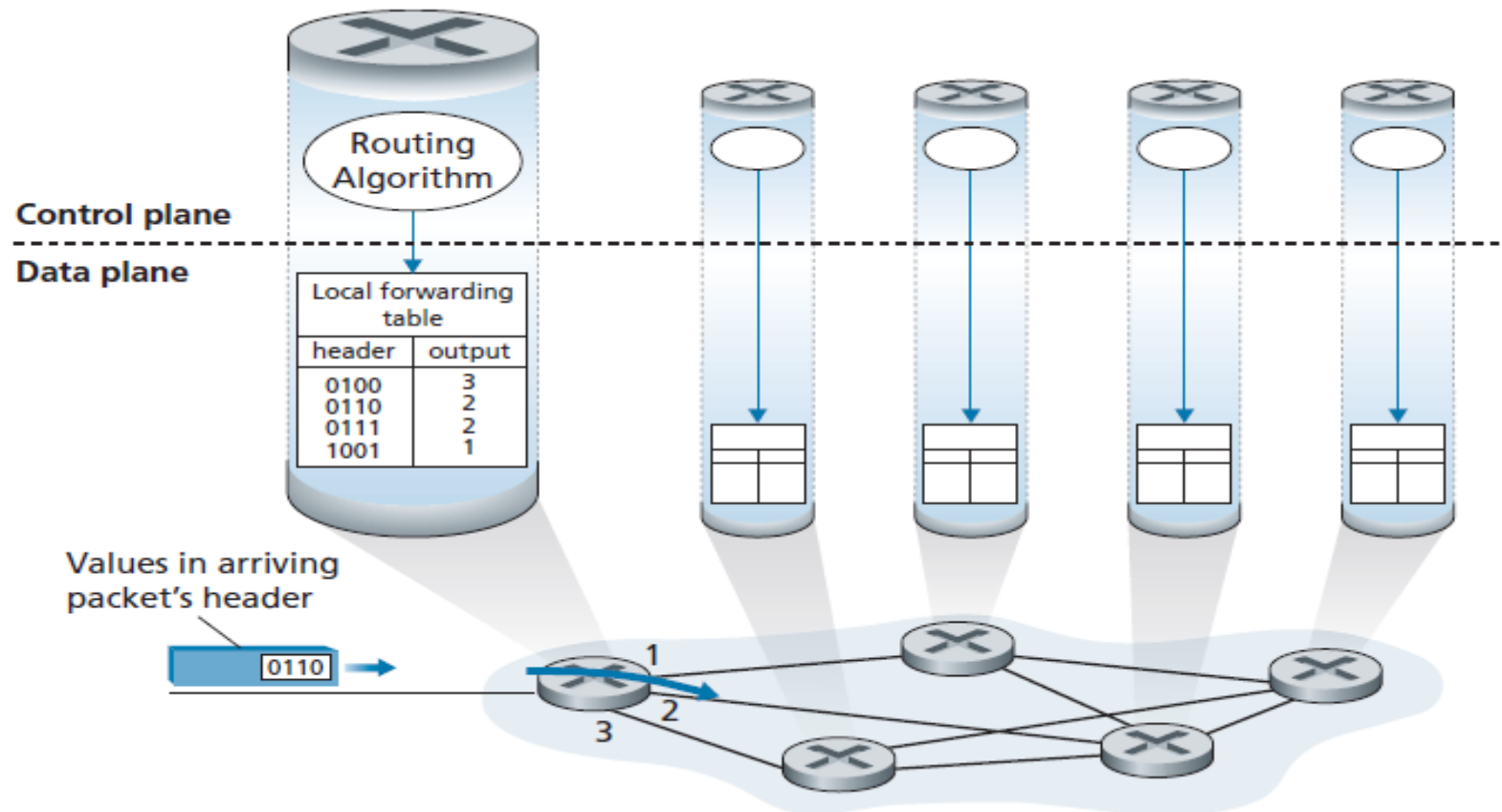
Example-2

- **E-commerce (e.g., order delivery)**
- **Routing**
- Platform decides:
 - Which warehouse will fulfill order
 - Which courier partner to use
- **Forwarding**
- Warehouse staff:
 - Pack item
 - Send to courier hub
- Courier hubs:
 - Pass package along step-by-step

Forwarding Table

- A key element in every network router is its **forwarding table**
- A router forwards a packet by examining the value of one or more fields in the arriving packet's header
- Uses these header values to index into its forwarding table
- The value stored in the forwarding table entry for those values **indicates the outgoing link** interface at that router to which that packet is to be forwarded

Forwarding Table



Routing Algorithm determines values in forwarding table

Ex-1

- A router has the following directly connected networks:

Network Address	Subnet Mask	Interface
192.168.1.0	255.255.255.0	eth0
192.168.2.0	255.255.255.0	eth1
10.0.0.0	255.0.0.0	eth2

Determine which interface will be used for:

- 192.168.1.45
- 192.168.2.200
- 10.5.6.7

Interface = connection point of a device to a network

Ex-1-Soln

- 192.168.1.45
- Matches: 192.168.1.0/24

Network Address	Subnet Mask	Interface
192.168.1.0	255.255.255.0	eth0
192.168.2.0	255.255.255.0	eth1
10.0.0.0	255.0.0.0	eth2

Ex-1-Soln

- **192.168.1.45**
- Matches: 192.168.1.0/24
- Interface → **eth0**

Ex-1-Soln

- **192.168.1.45**
- Matches: 192.168.1.0/24
- Interface → **eth0**
- **192.168.2.200**
- Matches: ?

Ex-1-Soln

- **192.168.1.45**
- Matches: 192.168.1.0/24
- Interface → **eth0**
- **192.168.2.200**
- Matches: 192.168.2.0/24
- Interface → ?

Ex-1-Soln

- **192.168.1.45**
- Matches: 192.168.1.0/24
- Interface → **eth0**
- **192.168.2.200**
- Matches: 192.168.2.0/24
- Interface → **eth1**
- **10.5.6.7**

Ex-1-Soln

- **192.168.1.45**
- Matches: 192.168.1.0/24
- Interface → **eth0**
- **192.168.2.200**
- Matches: 192.168.2.0/24
- Interface → **eth1**
- **10.5.6.7**
- Matches: 10.0.0.0/8
- Interface → **eth2**

Routing Algorithm

- A routing algorithm runs in every router
- Both forwarding and routing functions are contained within a router
- The routing algorithm function in one router communicates with the routing algorithm function in other routers to compute the values for its forwarding table
- Routing → Sending packets from source to destination

Forwarding Table

- Every switch/router maintains one in the network
- Switch/router looks up the destination in a table to see the destination and finds the link
- Table tells what to do with the packet
- Table → Routing Table
-

Routing

- Construction of routing tables
- At each switch/router
- Done through SW

Costs

- Show links with costs
- Delay or latency
- E.g. Dollar cost of shipping data charged by service providers
- This solves the problem of finding the minimum cost of routing the packet

Routing Table

- DB of 3 columns
 - Destination
 - Cost
 - Link to use (**next hop**) → route
-
- Path → sequence of links (A - C - E)

Routing Table

- Each Router maintains its own table
- Includes:
 - Destination
 - Cost / Distance
 - Next Hop

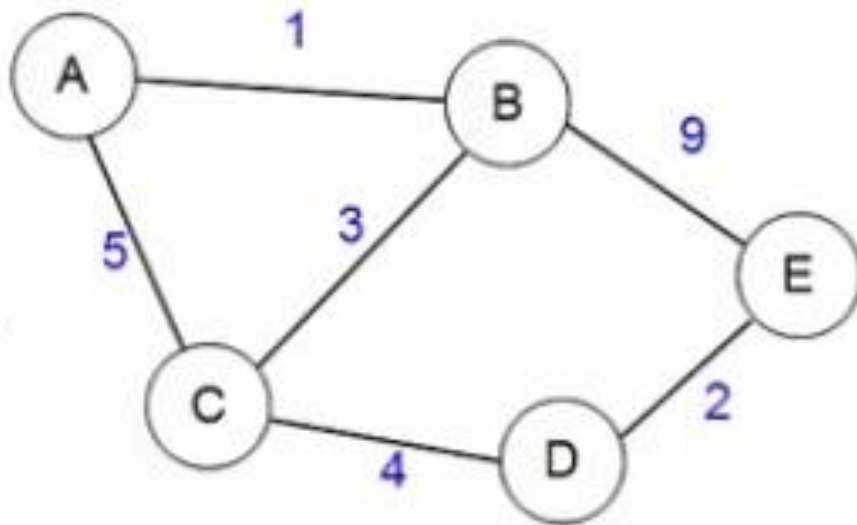
Distance Vector Routing

- Looks at the network as a set of nodes and edges → A graph
- Edge → link
- Each link has a cost

Distance Vector Routing

- Every node has info about immediate neighbor
- Uses Bellman Ford Equation
- Initially : Distance/cost to reach neighbor is known
- Finally : Distance/cost to all nodes are known
- Routing Table → Distance Vector
- Here Cost = estimated cost to destination bcz it is updated in each iteration

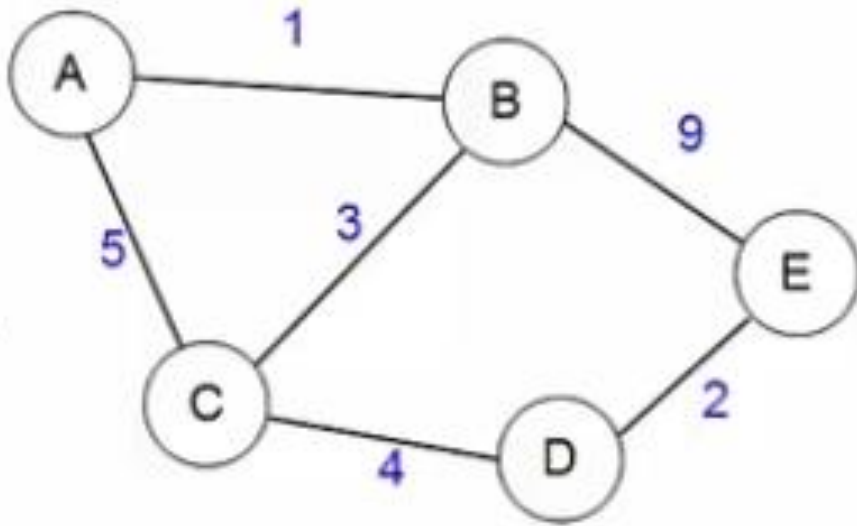
Given Network Graph



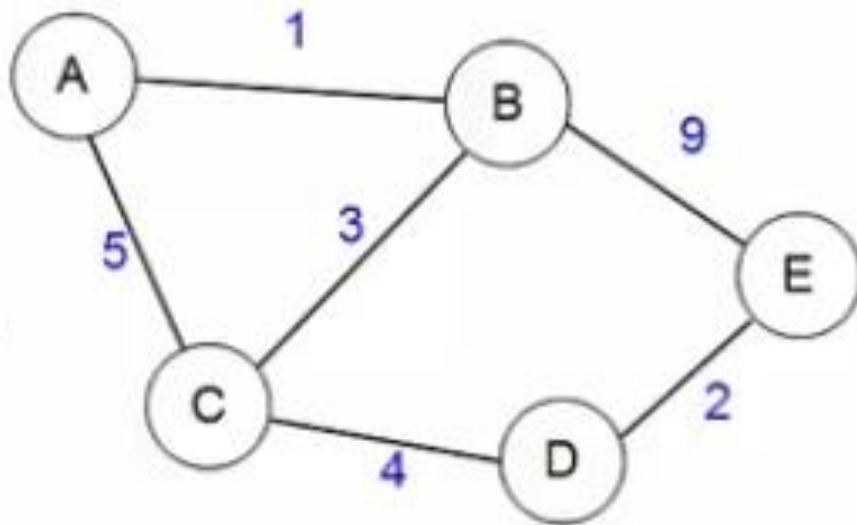
Initial Routing Table - Node A

	COST	
Destination	Distance	Next Hop
A	0	A
B	1	B
C	5	C
D	∞	-
E	∞	-

For Router B

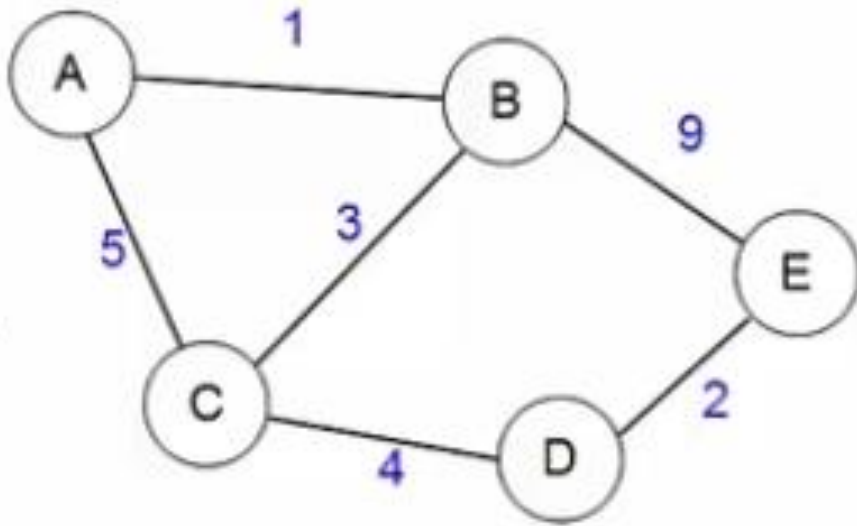


For Router B

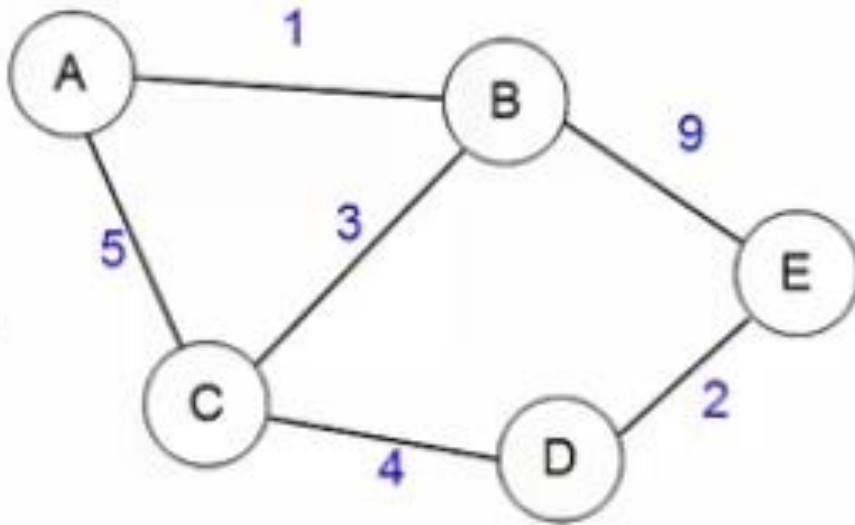


Initial Routing Table - Node B		
	COST	
Destination	Distance	Next Hop
A	1	A
B	0	B
C	3	C
D	∞	-
E	9	E

For Router/Node C

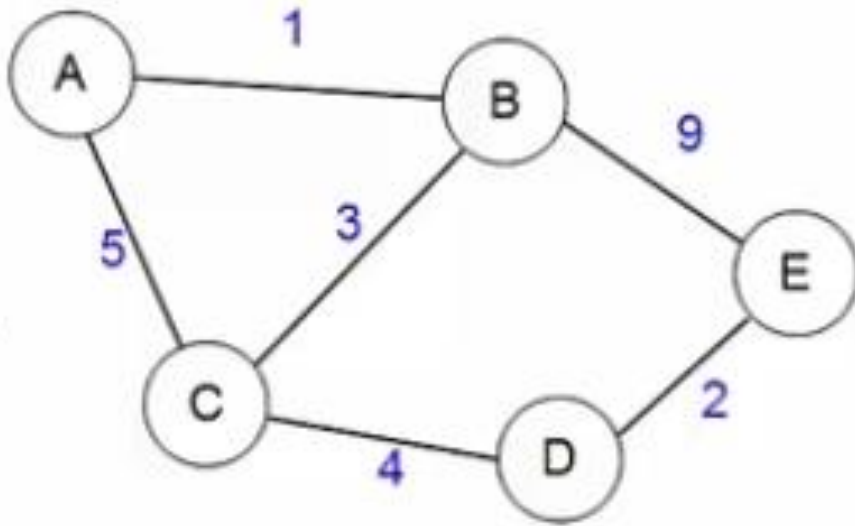


For Router/Node C

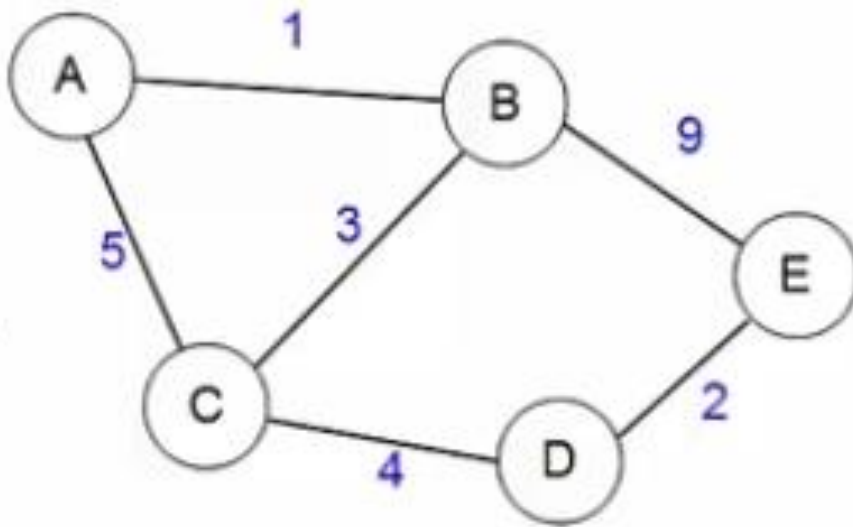


Initial Routing Table - Node C		
	COST	
Destination	Distance	Next Hop
A	5	A
B	3	B
C	0	C
D	4	D
E	∞	-

For Node D

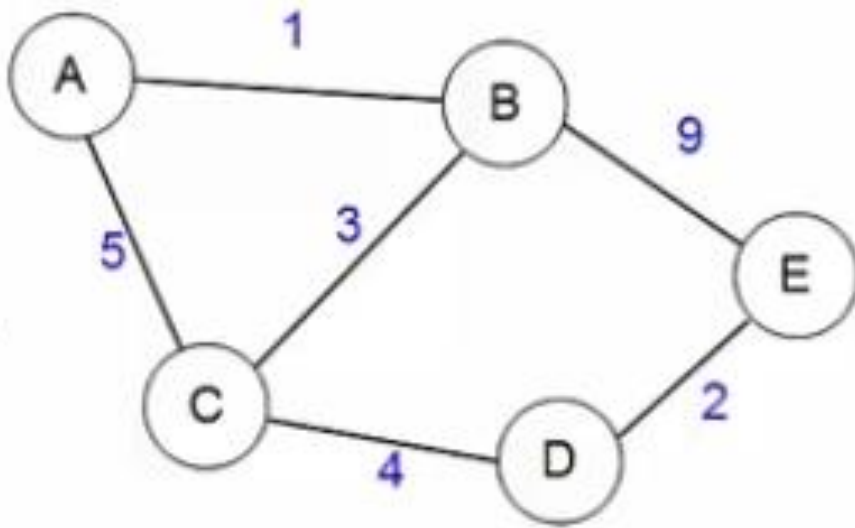


For Node D

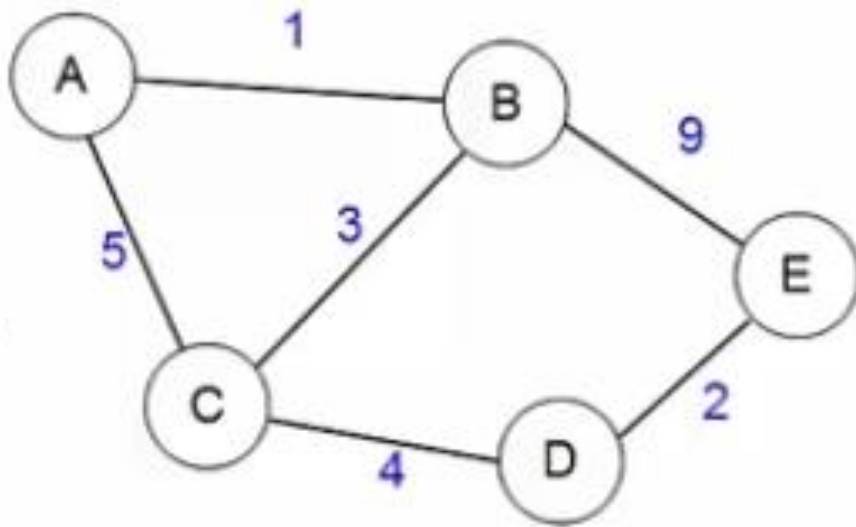


Initial Routing Table - Node D		
	COST	
Destination	Distance	Next Hop
A	∞	-
B	∞	-
C	4	C
D	0	D
E	2	E

For Node E



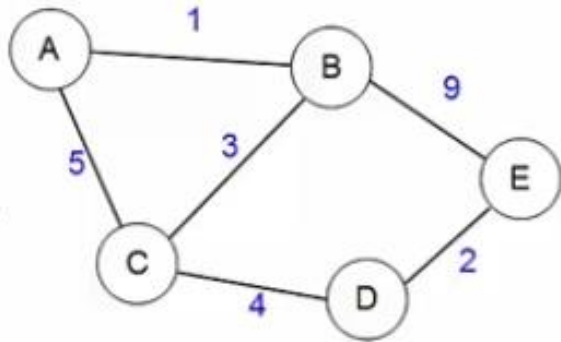
For Node E



Initial Routing Table - Node E		
	COST	
Destination	Distance	Next Hop
A	∞	-
B	9	B
C	∞	-
D	2	D
E	0	E

Node A

- Receives distance vectors of the directly linked neighbors



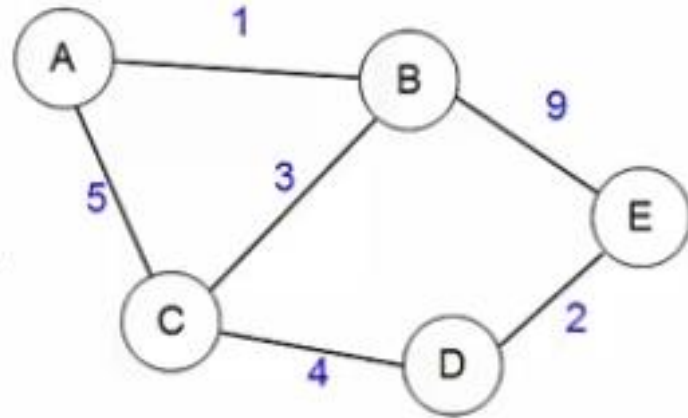
At Node A	
From B	From C
Distance	Distance
1	5
0	3
3	0
∞	4
9	∞

Initial Routing Table - Node B		
	COST	
Destination	Distance	Next Hop
A	1	A
B	0	B
C	3	C
D	∞	-
E	9	E

Initial Routing Table - Node C		
	COST	
Destination	Distance	Next Hop
A	5	A
B	3	B
C	0	C
D	4	D
E	∞	-

At Node A

- To reach A
- Distance = 0
- Next Hop = A
- To reach B
- Distance = $(A \rightarrow B + B \rightarrow B) = 1 + 0 = 1$
- Next Hop = B
- To reach C
- Distance = $(A \rightarrow C + C \rightarrow C) = 5 + 0 = 5$
- Next Hop = C

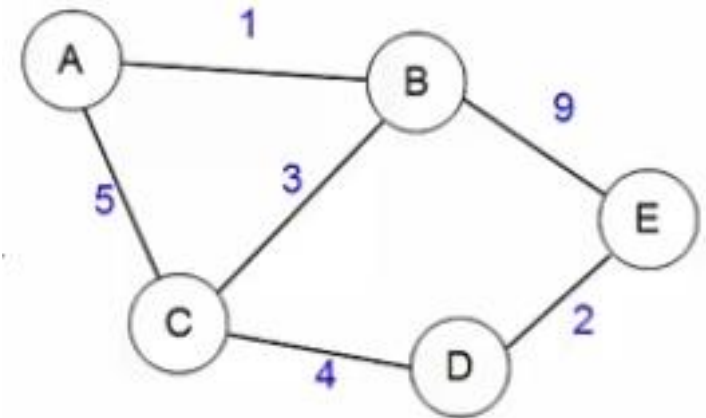


At Node A	
From B	From C
Distance	Distance
1	5
0	3
3	0
∞	4
9	∞

At Node A

- To reach D
- Using B
- Distance = $(A \rightarrow B + B \rightarrow D) = 1 + \infty = \infty$
- Using C
- Distance = $(A \rightarrow C + C \rightarrow D) = 5 + 4 = 9$
- **MINIMUM = 9**
- Next Hop = C,D

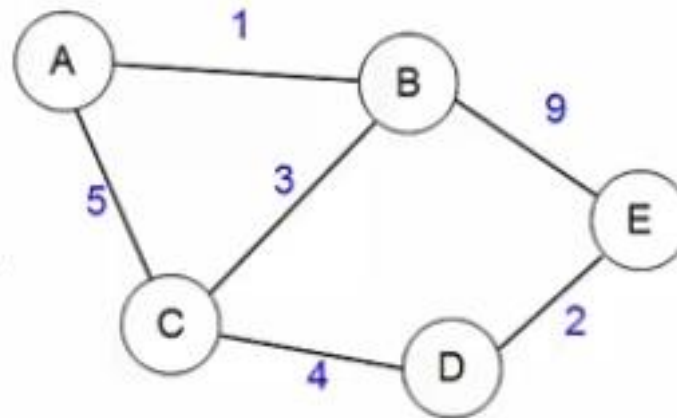
At Node A	
From B	From C
Distance	Distance
1	5
0	3
3	0
∞	4
9	∞



At Node A

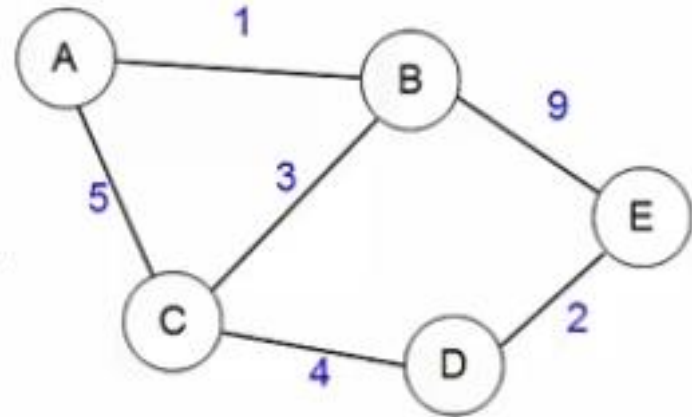
- To reach E
- Using B
- Distance = $(A \rightarrow B + B \rightarrow E) = 1 + 9 = 10$
- Using C
- Distance = $(A \rightarrow C + C \rightarrow E) = 5 + \infty = \infty$
- **MINIMUM = 10**
- Next Hop = B,E

At Node A	
From B	From C
Distance	Distance
1	5
0	3
3	0
∞	4
9	∞



Updated routing table at A

Initial Routing Table - Node A		
	COST	
Destination	Distance	Next Hop
A	0	A
B	1	B
C	5	C
D	∞	-
E	∞	-



Updated Routing Table - Node A		
	COST	
Destination	Distance	Next Hop
A	0	A
B	1	B
C	5	C
D	9	C,D
E	10	B,E

Class Practice: work out the updated Routing Tables for each node

In Class Task

- Update the routing tables at D and E using Paper and pen
- Upload the Pdf after scanning your hand done work (may use CamScanner) in GCR under Assignment 3
- Deadline will be strictly followed

Book-References

- Chapter 18,8 - Douglas E. Comer